

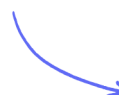
Multiple Choice Questions

Redox Chemistry & Acid-Base Titrations

Oxidation Numbers - Introduction / Oxidation & Reduction / Disproportionation Reactions / Ionic Equations / Acid-Base Titrations with Indicators

Easy (3 questions)	/3
Medium (13 questions)	/16
Hard (1 question)	/3
Total Marks	/22

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Easy Questions

1 What is the oxidation number of chromium in $\text{Na}_2\text{Cr}_2\text{O}_7$?

- A. +1
- B. +2
- C. +3
- D. +6

(1 mark)

2 In an oxide of nitrogen, the oxidation number of nitrogen is +4.

Which is the formula of the oxide?

- A. N_2O
- B. N_2O_3
- C. N_2O_4
- D. N_2O_5

(1 mark)

3 What is the oxidation number of phosphorus in the phosphate ion, PO_4^{3-} ?

- A. -3
- B. +3
- C. +5
- D. +7

(1 mark)

Medium Questions

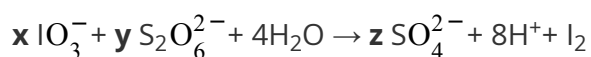
- 1 Hydrogen peroxide, H_2O_2 , breaks down into water and oxygen.

In terms of oxidation and reduction, how do hydrogen and oxygen change in this reaction?

		Hydrogen	Oxygen
☐	A	oxidised	reduced
☐	B	oxidised and reduced	unchanged
☐	C	reduced	oxidised
☐	D	unchanged	oxidised and reduced

(1 mark)

- 2 Iodate(V) ions, IO_3^- , oxidise dithionate ions, $\text{S}_2\text{O}_6^{2-}$, according to the equation

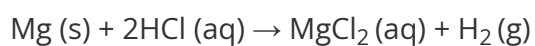


What are the balancing numbers **x**, **y** and **z**?

		x	y	z
☐	A	2	1	2
☐	B	2	2	4
☐	C	2	5	5
☐	D	2	5	10

(1 mark)

3 Magnesium reacts with hydrochloric acid.



Which statement about this reaction is correct?

- A.** magnesium atoms act as oxidising agents
- B.** hydrogen molecules act as reducing agents
- C.** hydrogen ions act as oxidising agents
- D.** chloride ions act as oxidising agents

(1 mark)

4 The formulae of four ions are shown.

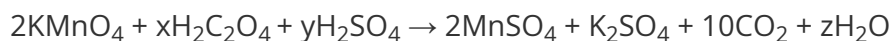
Formula of ion
CrO_4^{2-}
AlO_2^-
$[\text{Fe}(\text{CN})_6]^{4-}$
$[\text{CrCl}_2(\text{H}_2\text{O})_4]^+$

How many of these ions contain a metal with an oxidation number of +3?

- A.** one
- B.** two
- C.** three
- D.** four

(1 mark)

5 (a) This question is about the reaction shown.



What values of x, y and z are needed to balance the equation?

		x	y	z
☐	A	5	6	8
☐	B	10	3	4
☐	C	5	3	8
☐	D	10	6	4

(1 mark)

(b) What is the reducing agent in the reaction?

- A.** H^+
- B.** $\text{C}_2\text{O}_4^{2-}$
- C.** MnO_4
- D.** SO_4

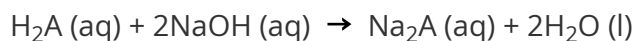
(1 mark)

6 Which name is correct for the ion SO_4^{2-} ?

- A.** sulfate(II)
- B.** sulfate(IV)
- C.** sulfate(VI)
- D.** sulfate(VIII)

(1 mark)

7 A diprotic acid, H_2A , was titrated with sodium hydroxide solution.



A 25.0 cm^3 portion of $0.100 \text{ mol dm}^{-3}$ sodium hydroxide solution required 12.80 cm^3 of the solution of the diprotic acid for complete neutralisation. What is the concentration of H_2A in mol dm^{-3} ?

A. 2.56×10^{-2}

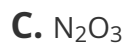
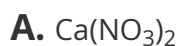
B. 9.77×10^{-2}

C. 1.95×10^{-1}

D. 3.91×10^{-1}

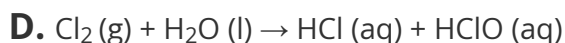
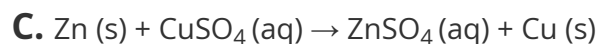
(1 mark)

8 In which compound is the oxidation number of nitrogen +5?



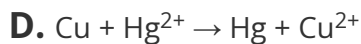
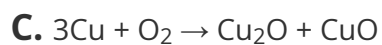
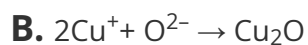
(1 mark)

9 Which reaction is **not** a redox reaction?



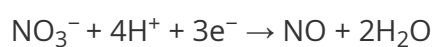
(1 mark)

10 In which reaction is the copper species acting as an oxidising agent?

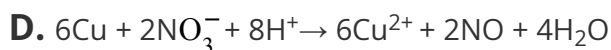
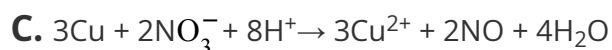
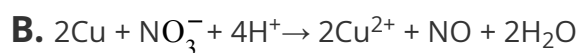
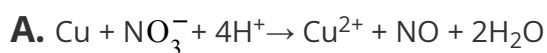


(1 mark)

11 Two half-equations for a reaction are shown.



What is the overall ionic equation for this reaction?



(1 mark)

12 A titre of 13.25 cm^3 was obtained using a 50 cm^3 burette.

What is the percentage uncertainty in the titre?

[Each reading of the burette has an uncertainty of $\pm 0.05 \text{ cm}^3$]

A. $\pm 0.38\%$

B. $\pm 0.75\%$

C. $\pm 1.5\%$

D. $\pm 7.5\%$

(1 mark)

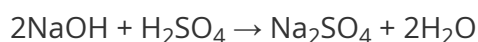
13 (a) a)

A pellet of sodium hydroxide has a mass of 0.700 g. Some pellets were dissolved to make 350 cm³ of 0.25 mol dm⁻³ solution. [*M_r* value: NaOH = 40] How many pellets were dissolved?

- A.** 4
- B.** 5
- C.** 8
- D.** 125

(1 mark)

- (b)** 25.0 cm³ of the sodium hydroxide solution prepared in (a) was placed in a conical flask and titrated with sulfuric acid.



Calculate the number of moles of sulfuric acid that reacted.

- A.** 0.0031
- B.** 0.0063
- C.** 0.013
- D.** 0.044

(1 mark)

- (c)** Phenolphthalein indicator was used for the titration in (b).

What was the colour change at the endpoint?

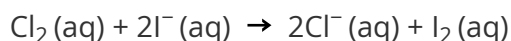
- A.** colourless → pink
- B.** pink → colourless
- C.** orange → yellow
- D.** yellow → orange

(1 mark)

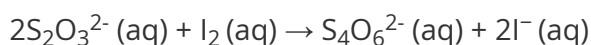
Hard Questions

- 1 (a)** A group of students carry out an experiment to find the concentration of chlorine, Cl_2 (aq), in a solution.

Excess potassium iodide solution is added to a 10.0 cm^3 sample of the chlorine solution.



The iodine produced is titrated with a solution of thiosulfate ions of known concentration, using starch indicator.



The concentration of the Cl_2 (aq) is between 0.038 and $0.042 \text{ mol dm}^{-3}$.

What concentration of thiosulfate ions, in mol dm^{-3} , is required to give a titre of approximately 20 cm^3 ?

- A.** 0.010
- B.** 0.020
- C.** 0.040
- D.** 0.080

(1 mark)

- (b)** What is the most suitable volume of 0.1 mol dm^{-3} potassium iodide solution, in cm^3 , to add to the 10.0 cm^3 of chlorine solution?

- A.** 7.6
- B.** 8.0
- C.** 8.4
- D.** 10.0

(1 mark)

(c) What is the colour change at the end-point of the titration?

- A.** colourless to pale yellow
- B.** pale yellow to colourless
- C.** colourless to blue-black
- D.** blue-black to colourless

(1 mark)